

pces Internals

(last update August 14, 2025)

CmpPtnMsg

As we have outlined in [Introduction to pces](#), the **pces** tool expresses patterned computations in terms of functions that receive messages as input, and produce messages as output. The simulation of a function is triggered by the receipt of a message directed to that specific function. A field of that message, *MsgType*, is of particular importance. A function instance may have a number of code methods to use in responding to a message, and the value of *MsgType* is used to look up which of those methods is called in response.

For clarity and reference it is worthwhile discussing the data structure describing inter-function messages.

```
type CmpPtnMsg struct {
    ExecID    int    // initialize when with an initating comp pattern message.
    FlowID    int    // identity of flow when message involves flows
    MsrID     int    // when carrying a measure, the identity of that measure
    PrevCPID  int    // ID of the previous comp pattern
    PrevLabel string // label of the previous function

    XCPID  int    // cross-CP id
    XLabel string // cross-CP label
    XMsgType string // cross-CP message type
    CPID   int    // identity of the destination comp pattern
    Label  string // label of the destination function

    MsgType string // describes function of message
    MsgLen  int    // number of bytes
    PcktLen  int    // parameter impacting execution time

    MsrSrtID  int    // ID of measurement function ID that started measurement
    StartMsr  float64 // time when measurement started

    RtnCPID  int    // used on service calls
    RtnLabel  string // used on service calls
    RtnMsgType string

    Rate      float64 // when non-zero, a rate limiting attribute
    FlowState string // "srt", "end", "chg"
    NetLatency float64
    NetBndwdth float64
    NetPrLoss  float64
    Payload    any // free for "something else" to carry along
}
```

A unique *ExecID* value is established with the creation of a first message to pass through an execution thread. Any message that passes through a function carries the same *ExecID* on the way out as it did on the way in. This makes it a useful index for marking messages associated with the same execution thread across functions.

pces and **mrnes** support a network object called a *Flow*, which is traffic whose description is given in terms of average bitrate rather than a particular packet. A *CmpPtnMsg* can carry information about a *Flow*, in particular an edge where a bitrate change is propagated through the network. Flows have identities, *FlowID* is a unique id given when a flow is created. Both flows and messages get different levels of priority when competing for resources, the *ClassID* identifies the priority class. The larger the class id, the greater the priority.

Figures 1 and 2 illustrate computation patterns that are a chain of functions. The second function executed, here labeled 'srtMeasure', begins a measurement, while 'endMeasure' completes the measurement. To correlate measurements on the same execution thread across different measurement points, when a measure is first started the identity of the measurement point is written into the message (field *MsrSrtID*), and like the *ExecID* is preserved through function calls.

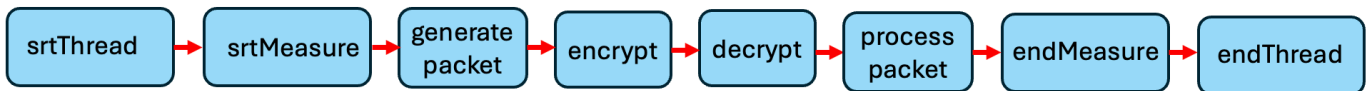


Figure 1: Chain of **pces** functions

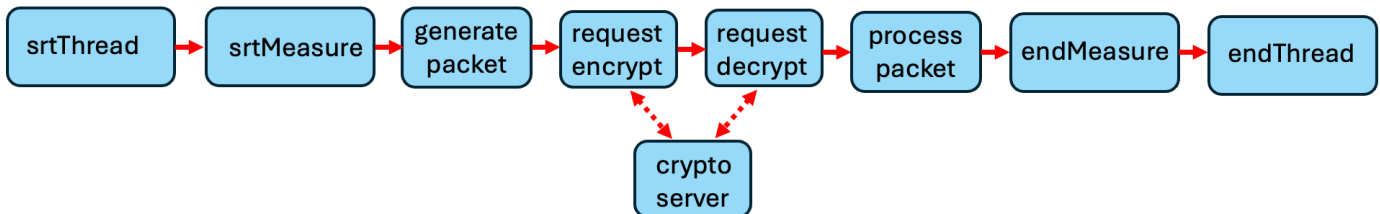


Figure 2: Chain of **pces** functions including server

It is sometimes useful for a function to know the source of a message it is processing. Therefore the *PrevCPID* and *PrevLabel* fields identify the *CmpPtn* of the source function, and its label within that *CmpPtn*.

XCPID and *XLabel* are fields available to user code to specify 'the next' *CmpPtn* and function Label to visit when the message encounters the next function of the *transfer* class. Their presence was driven by a use case during development where we wished to bind the target of a message that crosses *CmpPtn*'s to a message at the point the message is first created.

MsgType is a critical field, driving the response of a function to receipt of a message. The field will usually be overwritten as the message passes from function to function.

MsgLen carries the number of bytes in the *CmpPtn* frame, and *PcktLen* carries the number of bytes in the data frame, the latter being a parameter used when the execution time of a function processing the message is determined.

MsgSrtID is an ID for a measurement that was started earlier in the execution thread this message is part of. *StartMsr* is the simulation time when that measurement started.

RtnCPID, *RtnLabel*, and *RtnMsgType* are fields set by a function of the *srvReq* class, that requests service from some server that is typically in a different *CmpPtn*. After service is provided there, the message is returned to the *CmpPtn* and function indicated by the first two of these fields, with a *MsgType* field given by in message's *RtnMsgType* field. These fields are involved in the communication from 'request encrypt' to 'crypto server' in Figure 2.

When the message is associated with a *Flow*, the *Rate* field gives the bitrate of the edge of the flow described in the message. *FlowState* is a code that indicates whether the message describes the leading edge of a new *Flow*, the trailing edge of a flow that is disappearing, or a rate change to a flow that has already been established and it not now terminating (even if the *Rate* drops to 0.0).

NetLatency, *NetBndwdth*, and *NetPrLoss* carry information about the latency, minimal bandwidth, and estimated probability of packet loss observed as a message traverses its execution thread.

Finally, a *CmpPtnMsg* struct can carry a pointer along the execution thread to any structure at all, in the *Payload* field. For example, it may be that the source of a message to deliver through function calls and network traversals comes from an external emulator, or even, real external device. That network frame can be attached to a *CmpPtnMsg* through *Payload* and carried all the way to the destination of the execution thread where it can be accessed, and passed along to another real device or emulator. There may be other applications, e.g. the structure *Payload* points to describes an event from a different simulator that is using **pces/mrnes** to simulate a connection between endpoints in the external simulator.

pces Function Organization

The input/output relationships between functions can be expressed by thinking of the functions as nodes in a graph, and defining a directed edge from one function to another when a message leaving the source function *may* be presented as input to the destination function. We emphasize *may* here rather than *will* because there is nothing in the **pces** semantics that requires the transfer when an edge is defined in the **pces** model expression of topology. That said, in the default message handling functions provided in the github.com/iti/pces repository, with one exception a function has at most one output edge. We document how users can augment **pces** model expression by writing their own message-processing methods in [User Extensions to pces](#), and these may well generate multiple outputs from a single function execution.

A particular **pces** input file (nominally called 'cp.yaml') defines the *CmpPtn*'s, identifies the functions assigned to them with their 'labels', and identifies known input/output relationships between functions though the definition of 'edges'. Each edge is labeled with the *MsgType* of the message that may pass from one function to the other.

Simple *CmpPtns* can be a chain of functions, just as in Figure 1. In Figure 2 we could have the crypto server reside in a *CmpPtn* different from the other functions, or the same. There is a certain coherence in presentation if functions in multiple *CmpPtns* request service from it, it then has no particular binding to any one of them. As suggested by the explanation that the crypto server may be in a different *CmpPtn* than the functions that request its service, communication between functions in different *CmpPtns* is possible.

Processing/Routing

The logic of delivering messages from one Func to another has two layers. At the application layer the routing is governed by the information applied to Func outbound messages. When a message is presented to a Func, **pces** determines what computational logic will be applied. The message delivery is associated with an edge directed from one Func to the recipient, and combination of source Func identity (both its label and the computational pattern that holds it) and message type on that edge define the method code to use. This mapping is inferred at start-up time by analysis of all the *CompPatternEdge* and *XCPEdge* dictionaries defined by the model in *cp.yaml*.

To handle the message receipt, **pces** scheduled execution of an event (*EnterFunc*) that starts the execution of every Func response, and provides the identity of the Func receiving the message, and the particular message being delivered. *EnterFunc* looks up the method code, and uses this to look up the method-code-specific routine that starts the message processing. This routine may analyze and alter the Func's internal configuration and state, and use the results of that to look up an execution time requirement. It then requests from the hosting *Endpt* CPU that amount of service, and in the request records the specific method code, message, and identity of the event handler to schedule to report the service request completion.

When the service request completion handler executes it handles generation of messages that model the Func's response. It may generate any number of messages, including none. For each message it selects one of the Func's egress edges to 'carry' it to the next Func, and notes the destination Func's computation pattern and label on the message. Finally, it places all the response messages in a data structure accessible outside of the event handler, and schedules the **pces** event handler *ExitFunc* to execute immediately.

The corresponding *ExitFunc* execution acquires the list of responding messages. For each it determines whether the destination Func resides on the current *Endpt* CPU, and if so schedules the *EnterFunc* routine to execute immediately. Here then the model edges define application level routing within an *Endpt*. If it happens that the destination Func resides on a different *Endpt*, then *ExitFunc* hands the message off to the *mrnes* network simulator. The handoff request names the identities of the present and destination *Endpts*, and the global identity of the destination Func. It also names the **pces** routine *ReEnter* for the *mrnes* simulator to schedule when the message has traversed the network and arrived at its destination *Endpt*.

The network route the message takes and the accrued simulation time are all governed by the **mrnes** model. At present **mrnes** finds the shortest path between source and destination *Endpts*, where the waypoints on the path are switches and routers, and edges in the network graph are defined by the declared connections between device interfaces (see 3.6). The cost of every edge is assumed to be 1. *mrnes* discovers paths on a per-demand basis, and caches the results of the calculation so (at the cost of memory) the path between any given source and destination *Endpt* is done at most once.

pces Function Execution

The processing associated with executing a function in response to a message is largely the same for every function class. An event handler for a common function (*pces/cpg.go:EnterFunc()*) is scheduled, by a scheduling call that includes a pointer to a specific function instance where the execution is to occur, and a message to be passed to the code identified to execute the function. When *EnterFunc* executes it extracts the *MsgType* field from the message, and consults a dictionary (*msg2mc*) defined in the to-be-executed function's configuration information a 'method code'. The method code in turn is used as a key to a dictionary (*RespMethods*) that is specific to the function's class, to identify a 'response subroutine' which is then called to handle the message arrival. Parameters to that subroutine include a pointer to the specific function (and its data), the method code, and the message. The subroutine called is tailored to the behavior of the function's class and its data. Depending on the function, it constructs a code to use in looking up the elapse of simulation time associated with the function, and (eventually) modify the message, choose a destination for it, and initiate a process that leads to the delivery of the message to the destination, which is tantamount to scheduling *pces/cpg.go:EnterFunc()* again, with a different set of arguments.

Figure 3 illustrates these relationships.

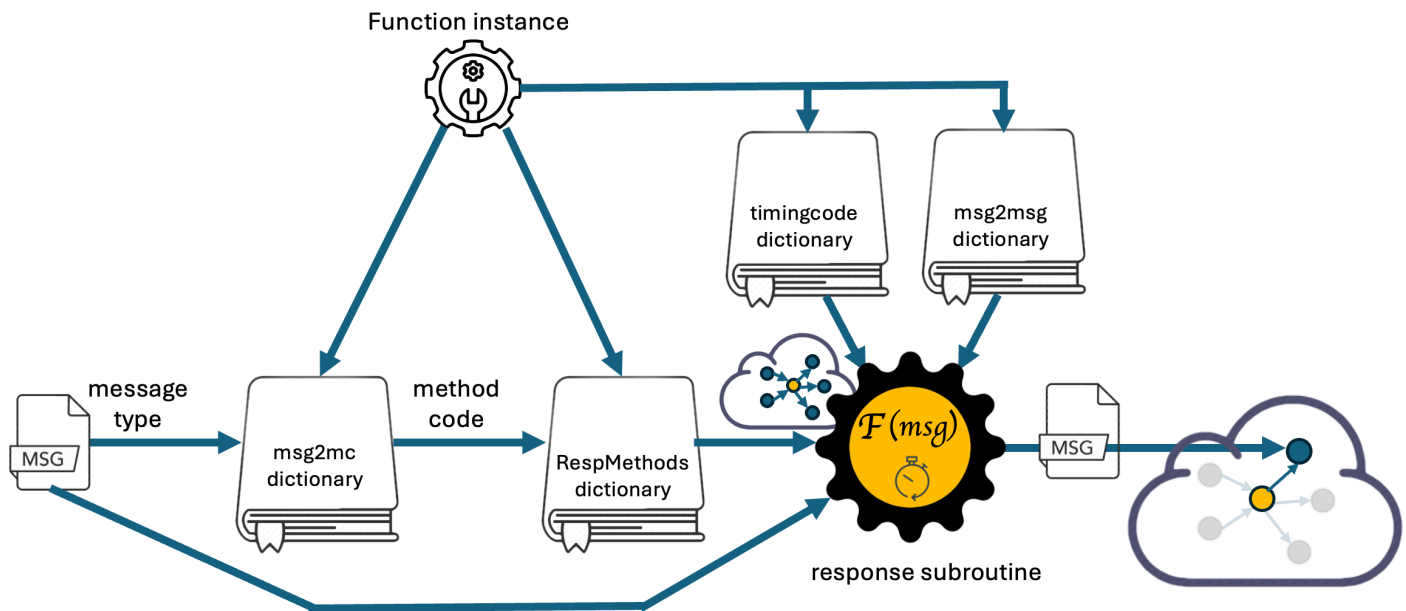


Figure 3: Selection and operation of response function for a given function instance.

Given that there will be users developing code for integration into **pces** models and that that integration sometimes involves scheduling calls to *EnterFunc()*, it is worthwhile to take a closer look at its code.

```

1 func EnterFunc(evtMgr *evtm.EventManager, cpFunc any, cpMsg any) any {
2     cpfi := cpFunc.(*CmpPtnFuncInst)
3
4     var stdFunc bool
5     methodCode := "default"
6     var cpm *CmpPtnMsg
7     var msgType string
8
9     if cpMsg != nil {

```

```

10     cpm, stdFunc = cpMsg.(*CmpPtnMsg)
11     if stdFunc {
12         msgType = cpm.MsgType
13     } else {
14         msgType = *cpMsg.(*string)
15     }
16     if len(msgType)>0 && len(cpfi.Msg2MC)>0 && len(cpfi.Msg2MC[msgType]) > 0 {
17         methodCode = cpfi.Msg2MC[msgType]
18     } else if len(msgType)>0 && len(cpfi.Msg2MC)>0 && len(cpfi.Msg2MC["*"])>0 {
19         methodCode = cpfi.Msg2MC["*"]
20     } else {
21         methodCode = "default"
22     }
23 }
24 if cpm != nil {
25     AddCPTrace(TraceMgr, cpfi.Trace, evtMgr.CurrentTime(), cpm.ExecID,
26         cpfi.ID, FullFuncName(cpfi,"EnterFunc"), cpm)
27 }
28
29 methods, present := cpfi.RespMethods[methodCode]
30 if !present {
31     classMethods, classpresent := ClassMethods[cpfi.Class][methodCode]
32     if !classpresent {
33         panic(fmt.Errorf("error class %s response method lookup code %s",
34             cpfi.Class, methodCode))
35     }
36     methods = &classMethods
37     cpfi.RespMethods[methodCode] = methods
38 }
39 // if the function is one that adds time (srvRsp, processPckt) then log
40 // it on an MsrRoute if the message is on one
41 if (cpfi.Class == "srvRsp" || cpfi.Class == "processPckt") && cpm.MsrSrtID > 0 {
42     MsrAppendID(cpm.ExecID, cpfi.ID)
43 }
44
45 methods.Start(evtMgr, cpfi, methodCode, cpm)
46 return nil
47 }

```

Figure 4: Code body of EnterFunc()

In line 1 we see the input arguments. Type 'any' in Go means, literally, any pointer, which means that to use a variable of that type one needs to know the type of the pointer it holds, and do a cast. Line 2 therefore is understood to cast the 'context' argument to be a pointer to a data structure associated with a specific function. Argument *cpMsg* is another 'any' variable, which by our convention is either empty (nil) or a pointer to a struct description a message passed between functions.

Lines 4 through 7 declare and initialize local variables.

One can schedule *EnterFunc()* to run and provide 'nil' (the empty pointer) for the message pointer, *cpMsg*. The test for non-emptiness at line 9 leads to execution at line 24 and then immediately to execution at line 29. Here we see then that if *EnterFunc()* is scheduled with a nil pointer for the message, it will take the default method code (established at line 5). If instead *cpMsg* is not empty, the logic of lines 10-15 determine whether *cpMsg* is a pointer to a message struct, or a string. In the former case it takes the *MsgType* field of the message to be the message type, and in the latter case it takes the value of the referenced string. The non-string logic serves a corner base when the function being executed is the first in the execution thread, and there is no prior message.

Lines 16-22 work out what the method code is when a message is presented, with the logic that if the message actually has a message type (it is supposed to) and the function has a message-type-to-method-code dictionary (*msg2mc*) that is non-empty and has an entry whose key is the message type, then the method code is the dictionary value for that key. Alternatively, if the dictionary exists, does **not** have a entry for the message type key, but **does** have a wildcard key '*', then the method code is the dictionary value of the wildcard. If neither of these conditions holds, the the method code is defined to be *default*.

If it happens that *cpMsg* carried a pointer to a message, lines 25 and 26 register an entry in a trace file (if tracing is enabled, a test which is not viewable in this code).

Line 29 attempts to look up a struct that holds a pointer to the response function. The dictionary searched is one specific to the function. If the search is successful the pointer is written into variable `methods` and the variable *present* is set to true, otherwise *present* is false. If the method code is not found in the function's own *RespMethods* dictionary a check is made against a dictionary *ClassMethods* kept as a backup for all function for a class. If the method code isn't found as a key to this dictionary either a run-time abort is executed. If on the other hand a match is found in *ClassMethods*, the entry found is copied into the function's own dictionary (lines 36-37).

Lines 41 and 42 save the identity of the entered function on a list compiled for a measurement being taken along a path; only functions that advance simulation are included.

Reaching line 45, the *Start* entry of the found struct is understood to be a function with a signature of

```
func(*evtm.EventMgr, *CmpPtnFuncInst, *CmpPtnMsg)
```

The functions called through line 45 are so-called 'Entry' functions for the different function classes. Text description of their operation are given later when we speak to each class.

When the processing associated with a message is completed, the message is placed in a data structure for retrieval by control function *ExitFunc()* (*pces/cpg.go:ExitFunc*) and execution of *ExitFunc()* is scheduled at a time that reflects the end of the simulation time delay introduced by the function, if any. These steps are performed by the function called at line 39, or (depending on the function and class) by event handling functions that it causes to be executed when the function simulation has completed. The destination CmpPtn, function label, and message type for each response message will have been decided and embedded in the message.

ExitFunc is given a pointer to the function that is completing and a pointer to the original message that triggered the evaluation. It uses the latter in particular to acquire the *ExecID* of the message, as the messages created in response will all carry that same *ExecID* code, and pulls up that list. It enters a loop where for each resulting message it determines if the destination function is mapped to a processor other than the current one, and if so sets up and executes a transfer of that message to a network simulation, here, **mrnes**. Part of the transfer setup is to record the particulars of the destination function so that when the message arrives at the targeted processor, the message is handed back to **pces** logic, and it has the information needed to re-introduce the message into **pces** level processing by scheduling an execution of *EnterFunc()* at the destination function, with the carried message, to occur at the simulation time the message reached the target processor.

pces Function Classes

Now consider each of the **pces** function classes, the data expected of a user to configure their use, and a description of what the default response function does when called. It is worth noting the functional signature of all routines called to process an arriving message (called type *StartMethod* in the **pces** code base). One argument is a pointer to a data structure that manages event scheduling, another is a pointer to the function to be executed, another is a string, and the final one a pointer to the message being processed.

By way of explanation, in presentation of the function configuration structures all of the names are lower case. The API for the **pces** input files uniformly specify variable names as lower case. Yet in the discussion we present about those variables in the data structures we will give those names but with some letters capitalized. The reason for this is that the Go library we use to serialize dictionaries for file storage requires that we declare a lower case label for every dictionary key, and that lower case label is what is written into the file. The mapping from lower case label to something else is likewise described, and implemented when the files are read. Our approach has consistently been to make the lower case label simply by changing upper case letters in dictionary keys to lower case. An example of this is given below for the definition of the *CompPattern* struct.

```
// CompPattern is a directed graph that describes the data flow among functions that
// implement an end-to-end computation
type CompPattern struct {
    // a model may use a number of instances of CompPatterns that have the same CPTYPE
    CPTYPE string `json:"cptype" yaml:"cptype"`

    // per-instance name of the pattern template
    Name string `json:"name" yaml:"name"`

    // instances of functions indexed by unique-to-pattern label
    Funcs []Func `json:"funcs" yaml:"funcs"`

    Services map[string]funcDesc `json:"services" yaml:"services"`

    // description of edges in Pattern graph
    Edges []CompPtnGraphEdge `json:"edges" yaml:"edges"`

    // description of external edges in Pattern graph
    ExtEdges []XCPEdge `json:"extedges" yaml:"extedges"}`
```

```
}
```

A statement like

```
CPTYPE string `json:"cptype" yaml:"cptype"`
```

conveys that when the struct is serialized to yaml or json, the label in the serialized version is 'cptype' even though the label in the struct is not all lower case.

We turn now to a description of the data in each function classes' configuration struct. There are two configuration state variables that appear in every class. User code can create 'group' labels and assign these to individual functions, writing the group names into the 'groups list.' Group labels can be used to identify functions within user code, particularly at initialization, by addition group names to the 'groups' list. The other common variable is 'trace', an integer-coded Boolean (0 or 1) indicating whether executions of this function should be written to the trace file. The reason for the integer coding is to circumvent issues we found serializing structs containing variables of Boolean type. Integer representation avoids this complication.

class start

A table describing parameters that a user specifies for a start function is given below.

name	Data type	Explanation
pcktlen	integer	For messages involved in crypto, the block size in bytes
msglen	integer	The number of bytes in the network frame that carries the message
msgtype	string	A code defined by the user to indicate what should happen with the message at the next function
data	string	A placeholder to allow as-yet-unspecified parameters to be included. User provided code will need to interpret the string
starttime	float	The simulation time at which the message should be defined and passed to the function's output.
groups	list (str)	A list of names of user-defined 'groups' this function is assigned to
trace	int	When equal to 1 the executions of this function are included in the trace generated for the simulation run, otherwise not.

A start function's default subroutine is called *startEnter()* (in file *pces/class.go*). It creates a message (of type *CmpPtnMsg*), and sets its *PcktLen*, *MsgLen*, and *MsgType* fields to the values described above in its configuration structure. It also sets an 'execution thread ID' in field *ExecID*, which is maintained as the message traverses its chain of functions. Some bookkeeping is done, and the created message is placed in a list that control function

ExitFunc() will examine for forwarding. It then schedules *ExitFunc()* to execute with zero simulation delay, passing to it a pointer to this start function, and the created list.

class feed

A pces model may accept as input packets generated externally. These may be introduced as the source of CmpPtn, in much the same way that a start function is the source of a CmpPtn. A table describing parameters that a user specifies for a feed function is given below.

name	Data type	Explanation
srcIP	string	The format "IP:port" where IP it the internal IP assigned to some model interface, which acts as the ingress point for the feed's packets. The packet's own source IP address is declared elsewhere in the model to be mapped to this srcIP. If "port" is "*" than any feed packet with the given IP as the source will be directed to the function, otherwise, only packets that match both in IP and port specifications. Exactly one of 'srcIP' and 'dstIP' below may be non-empty.
dstIP	string	The format "IP:port" where IP it the internal IP assigned to some model interface, which acts as the ingress point for the feed's packets. The packet's own destination IP address is declared elsewhere in the model to be mapped to this dstIP. If "port" is "*" than any feed packet with the given IP as the source will be directed to the function, otherwise, only packets that match both in IP and port specifications. Exactly one of 'dstIP' and 'srcIP' above may be non-empty.
msgtype	string	A code defined by the user to indicate what should happen with the message at the next function
data	string	A placeholder for data that might be used by a user extension to the feed class. Otherwise undefined.
groups	list (str)	A list of names of user-defined 'groups' this function is assigned to
trace	int	When equal to 1 the executions of this function are included in the trace generated for the simulation run, otherwise not.

class measure

name	Data type	Explanation
msrname	string	Measurements are given a name, measurement identifiers are included on a message so that when a message passes through a measure function, measurement action is taken only if the measurement identifier on the message matches the measurement identifier associated with msrname.
msrop	string	The measurement operation to be applied. "start" initiates a new measurement, "end" completes one, and anything else simply records the measurement.
msg2mc	dictionary (str -> str)	msg2mc[msg.MsgType] gives a method code, which must be found as a key in the pces RespTable dictionary for the <i>measure</i> class that maps method codes to subroutines called to process the message arrival.
groups	list (str)	A list of names of user-defined 'groups' this function is assigned to
trace	int	If equal to 1 the executions of this function are included in the trace generated for the simulation run, otherwise not.

A measure function's default subroutine is called *measureEnter()* (in file *pces/class.go*). This routine examines the *msrop* code to see whether the function starts a measurement. If so it creates a data structure to track the measurement, gives it an integer tag, and copies that tag to a field in the incoming message. The default function assumes that the function has exactly one output whose complete specification (CmpPtn, function label, message type) have been specified by the **pces** *CmpPtn* graph. The message is so modified and is placed where *ExitFunc()* will deal with it.

class processPckt

name	Data type	Explanation
timingcode	dictionary (str -> str)	The message 'MsgType' field is used as a key to acquire an operation code used to index into the function execution time table and acquire the simulation delay associated with executing this function.
msg2mc	dictionary (str -> str)	msg2mc[msg.MsgType] gives a method code, which must be found as a key in the pces RespTable dictionary for the <i>processPckt</i> class that maps method codes to subroutines called to process the message arrival.
msg2msg	dictionary (str -> str)	msg2msg[msg.MsgType] gives a message type to use on the output message, in the event that there is more than one output edge associated with the function and so some choice must be made. The output edge chosen is the one whose message type matches msg2msg[msg.MsgType].
accelname	string	The presence of a non-empty accelname string tells pces to use the timings of a function accelerator card (e.g., hash function) on the processor to which the function is mapped, rather than the normal processor model.
groups	list (str)	A list of names of user-defined 'groups' this function is assigned to
trace	int	When equal to 1 the executions of this function are included in the trace generated for the simulation run, otherwise not.

The default subroutine for a *processPckt* function is in *pces/cgp.go*, and is called *processPcktEnter*. It first looks up the simulation time delay associated with this function, with this message. If it happens that accelname is not empty, it modifies the lookup to get timings based on the named accelerator's performance. The function next looks up the **mrnes** endpoint on which the function is executing, and calls the endpoint's **mrnes** scheduler to manage the advancement of simulation time, allowing for the inclusion of effects of competition for CPU resources by other concurrent processes. At such time as the scheduler knows when the service demands have been met, it schedules the execution of another of the function's event handlers, *processPcktExit*. Execution of *processPcktExit* marks the message with the destination CmpPtn, label, and message type based on the assumption of there being exactly one output, with the transfer particulars having been specified by the declaration of the relationships between **pces** functions.

class *srvReq*

name	Data type	Explanation
bypass	int	When 1 the message bypasses this function and continues directly to the next function in the chain. Provides an easy way to determine the impact on performance of including the service this function requests from a server, or not.
srvcP	string	The name of the computational pattern that contains the server which will be requested to provide the service.
srVlabel	string	The name of the specific function that will be called to provide the service
srVop	string	A code for the service operation to be requested from called server
rspop	string	A code for the operation that processes the server response, if any
msg2mc	dictionary (str -> str)	msg2mc[msg.MsgType] gives a method code, which must be found as a key in the pces RespTable dictionary for the <i>processPcket</i> class that maps method codes to subroutines called to process the message arrival.
msg2msg	dictionary (str -> str)	msg2mc[msg.MsgType] gives a method code, which must be found as a key in the pces RespTable dictionary for the <i>srVReq</i> class, that maps method codes to subroutines called to process the message arrival.
groups	list (str)	A list of names of user-defined 'groups' this function is assigned to
trace	int	When equal to 1 the executions of this function are included in the trace generated for the simulation run, otherwise not.

The default subroutine for *srVReq* functions is *pces/class.go/srVReqEnter*. It's first task is to determine the global location of the server to be queried for service. A first step is to determine whether that is specified by the expressed topology of the model's functions. If it happens that the topological (graph) description of the model's functions defines an edge from this function to some other function that is of class 'srVRsp', **pces** takes the destination function to be the server function. If the model function topology does not identify the server, *processPcketEnter* uses a non-empty value of srVCP to identify the server's source CmpPtn. These two scenarios are illustrated in Figure 3 below.

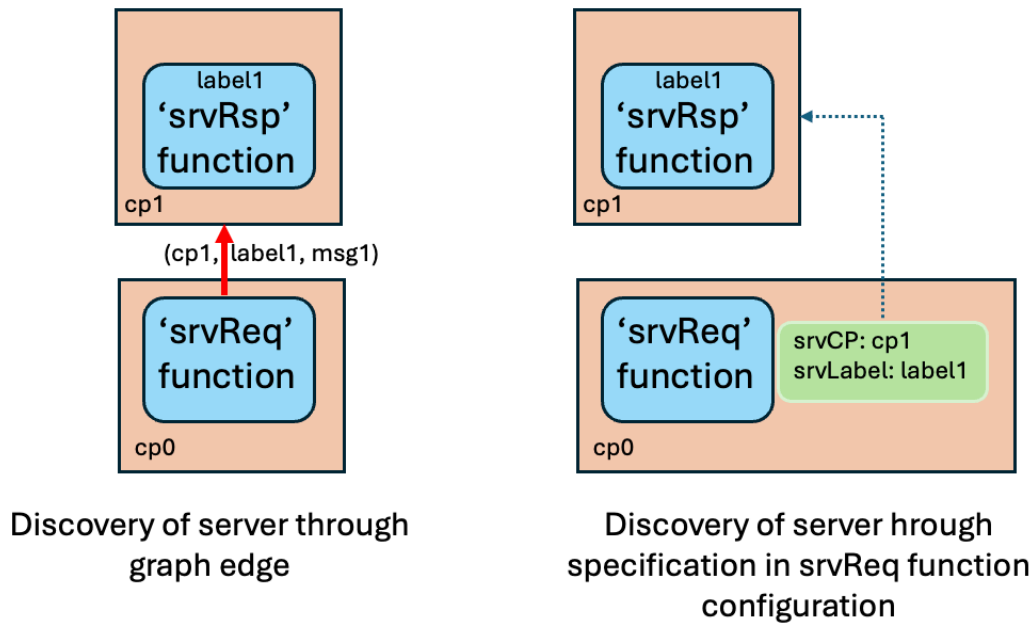


Figure 3: Direct methods for specifying server function that responds to `srvReq` Function query

There is another means of identifying the server function, that is driven by a use case where, in order to capture the dynamics of a 'zero trust' network architecture, every message that crosses processor boundaries must have the authenticity of the server checked. There are a variety of ways of implementing that in practice, one of which has the receiving entity issuing a challenge to the sender, e.g., sends a back a nonce that should be signed or encrypted and sent back to be checked. We can capture that pattern if the first function to be executed in response to a cross-processor message requests a service---authentication---from the sender. Expecting that executing the authentication computation would be provided in general, whenever needed, we imagine that that function resides potentially in a different *CmpPtn* than the sending function, and need not be the sending function itself. **pces** allows a *CmpPtn* definition to include table of 'Services', indexed by service op code, that points to the global identity of the function providing that service for that *CmpPtn*. We illustrate that below on the left in Figure 4. Without a direct use case but to complete the functionality, on the right in Figure 4 we illustrate that if the requesting function's `srvLabel` configuration variable is empty but its `srvCP` variable is not, then the same indirection logic can be used to identify the server, the only difference with the previous case being that the identity of the *CmpPtn* holding the Services table is specified in the configuration.

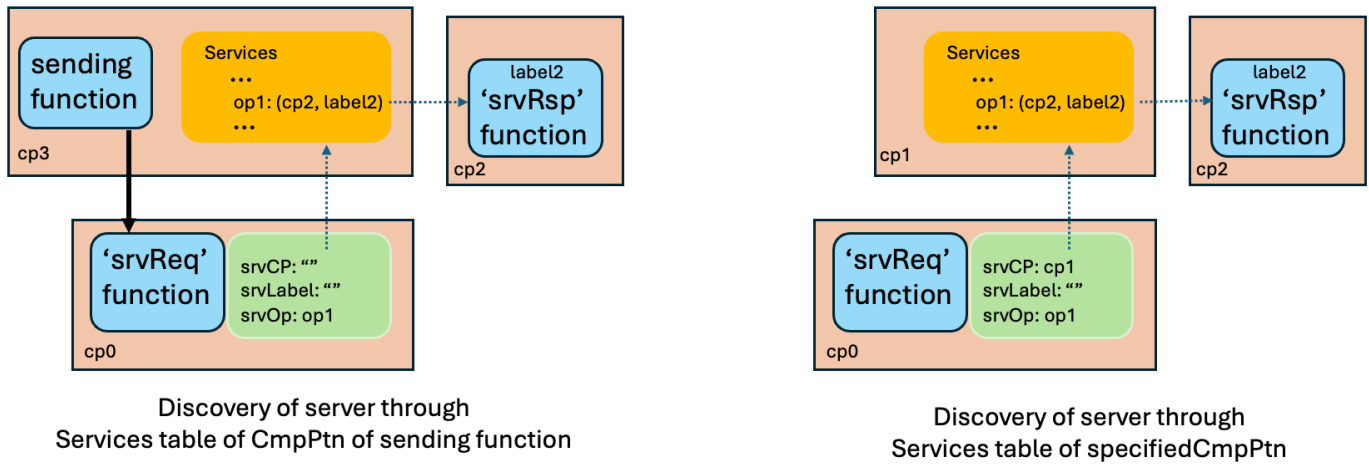


Figure 4: Indirect methods for specifying server function that responds to *srvReq* Function query

We have so far been deliberately circumspect in discussing the *srvOp* variable, because there is a bit of delicacy involved. Motivated by our known use case of determining performance sensitivity to different crypto algorithms and their parameters, we wish to configure the function that *requests* the crypto service with those parameters rather than the function that *provides* the service.. In this approach the requesting service just 'asks' the server to perform some particular operation, with that command carrying all the various parameters that impact the operation's execution time on the server.

The methodology **pces** adopts then is to have the *srvReq* function's configuration *srvOp* code follow a particular form. If that string has somewhere at least one '-' character, the substring defined from the first character up to the first '-' is a 'prefix' and is assumed to specify some service the server provides, like 'encrypt', 'decrypt', 'sign', 'auth', or 'hash'. Codes we've used in examples for crypto include 'encrypt-aes-256-CTR', 'decrypt-rc6-2048-CBC' which code the operation, crypto algorithm, key length, and cipher mode. The prefixes for these two examples are 'encrypt', and 'decrypt'. If the value of variable *srvOp* has a prefix, that prefix is the key used in a 'Service's dictionary lookup, otherwise the whole *srvOp* string is the key. The existence of a prefix involves the execution of a 'srvRsp' function also, as we will shortly see.

After the global identity of the server is acquired, *srvReqEnter* prepares the message it sends to it. That message includes fields that identify the *CmpPtn* and label of the requesting function, copy the calling function's *srvOp* code to be the message type, and crafts and stores in the message a field 'RtnMsgType' that should be used by the server as the message type of the message it returns to report service completion. *srvRspEnter* also includes an entry in the requesting function's 'Msg2MC' table which maps (as a key) the value crafted for the message 'RtnMsgType' field to the function *srvReqRtn*. When reporting the service completion, the server will use this code as the return message's *MsgType* field, which *EnterFunc* will use to look up subroutine *srvRspRtn* to deal with the response.

class *srvRsp*

name	Data type	Explanation
timingcode	dictionary (str -> str)	The message 'MsgType' field is used as a key to acquire an operation code used to index into the function execution time table and acquire the simulation delay associated with executing this function.
msg2mc	dictionary (str -> str)	msg2mc[msg.MsgType] gives a method code, which must be found as a key in the pces RespTable dictionary for the <i>srvRsp</i> class, that maps method codes to subroutines called to process the message arrival.
directprefix	list of strings	This list holds possible prefixes of service operations requested of the server.
groups	list(str)	A list of names of user-defined 'groups' this function is assigned to
trace	Int	When equal to 1 the executions of this function are included in the trace generated for the simulation run, otherwise not.

The default subroutine for *srvRsp* functions is *pces/class.go/srvRspEnter* . It understands the message type of the input message to carry a code for the service operation to be performed. *srvRspEnter* determines whether that code has a prefix, and if so, whether that prefix is found in the function's 'DirectPrefix' list. If a prefix is identified and is found in this list, the service op code is assumed to be an operation with a timing value in the function execution times table, and is used directly in looking up the operation's execution time. If instead the message's type does not have a prefix, or has a prefix that is not found in the function's directprefix table, *srvRspEnter* assumes that the service op code is a key found in the function's TimingCode dictionary, and uses the string to which the service op code is mapped as an op code that will be recognized in the function execution time table.

One way or the other the message type field of the message is transformed into an operation code used to index into the function execution time table, to retrieve the time delay associated with that operation on the processor to which the 'srvRsp' function is mapped. *srvRspEnter* next modifies fields of the received message to send it back to the 'srvReq' function that sent it; the identity of that function was recorded in the message when it was sent, as was a specially crafted value to use now as the message's MsgType field. *srvRspEnter* places the modified message where *ExitFunc* will find it, and finally schedules *ExitFunc* to execute after the time associated with the service operation execution elapses.

class metadata

There are situations where it is useful to associate meta data with a *CmpPtnMsg*, to be associated also with the network message that travels between endpoints in the midst of executing a *CmpPtn*. An example is when we want to model the additional cost of an add-on function in a router, e.g., adding a cryptographic signature or performing a deep packet inspection. In these cases we'd like to be able to apply this processing to some packets and not others, and can use meta data associated with the packet to recognize the distinction. Functions of the metadata class enable us to add and subtract meta data to a *CmpPtnMsg* that passes through it.

We characterize the meta data as a dictionary that maps strings to strings, which one can view as the name of a piece of meta data mapping to its value. **pces** has one built in meta data name, *compute*, which **mrnes** recognizes when simulating the passage of a packet through a router or switch. User extensions can expand this vocabulary.

A CmpPtnMsg message passing through a function of class metadata can have some existing items of meta data removed, and others added. The configuration is

Name	Data Type	Explanation
remove	List(string)	List of meta data names whose presence in the CmpPtnMsg are removed
metadict	Dictionary (str -> str)	Mapping of meta data variable names to their values
groups	list(str)	A list of names of user-defined 'groups' this function is assigned to
trace	Int	When equal to 1 the executions of this function are included in the trace generated for the simulation run, otherwise not.

On receipt of a CmpPtnMsg, the removal operations are applied first, followed by the additions. The CmpPtnMsg carries out message type as it had coming into the function.

class transfer

Default response method subroutines write into a 'CmpPtnMsg's 'MsgType' field a code that not only is used at the next function to identify what it will do, but also to route the message there. There is an implicit function-to-function routing step in Figure 3, as the message is depicted as leaving the response subroutine. Under the covers, the source function has a table of global addresses of functions to which it may direct the message. The 'MsgType' field is used to select the table entry, which holds the identity of the target's own CmpPtn and function label. Note then that a message can cross between CmpPtn's in the course of its execution thread.

A **pces** user can develop models where the target function may be selected by a function earlier in the execution thread, and for convenience **pces** provides another construction for transferring a message from one CmpPtn to another. This is the role of the 'transfer' function.

A 'transfer' functions configuration variables are listed below.

Name	Data Type	Explanation
carried	int	Integer-encoded Boolean indicating whether target identifies are carried by the message
xcp	str	Name of the target CmpPtn, if not carried in the message
xlabel	str	Label of the target function, if not carried in the message
xmsgtype	str	MsgType of the transferred message, if not carried in the message
msg2mc	str	msg2mc[msg.MsgType] gives a method code, which must be found as a key in the pces RespTable dictionary for the <i>svrRsp</i> class, that maps method codes to subroutines called to process the message arrival.
groups	list(str)	A list of names of user-defined 'groups' this function is assigned to
trace	int	When equal to 1 the executions of this function are included in the trace generated for the simulation run, otherwise not.

A 'transfer' function receives a message, looks up the global coordinates and message type of the message it pushes out, and hands the modified message off to **pces** to be routed like the output message of any other function. The target's global coordinates are either extracted from the function's configuration variables, or, if the 'carried' flag in the configuration is true, from the incoming message itself.